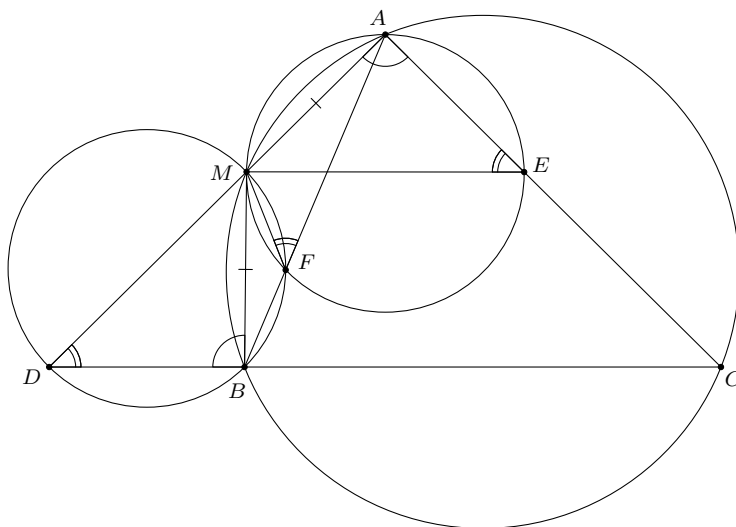


- Problem 57.** Let ABC be a triangle with $AB < AC < BC$. Suppose that
- M is the midpoint of $\widehat{AB}$ not containing C on the circumcircle of $\triangle ABC$;
 - the line AM meets the line BC at D ;
 - the line AB meets the circumcircle of $\triangle BMD$ again at F ; and
 - the line AC meets the circumcircle of $\triangle AMF$ again at E .

Prove that $BD = AE$.

Solution.



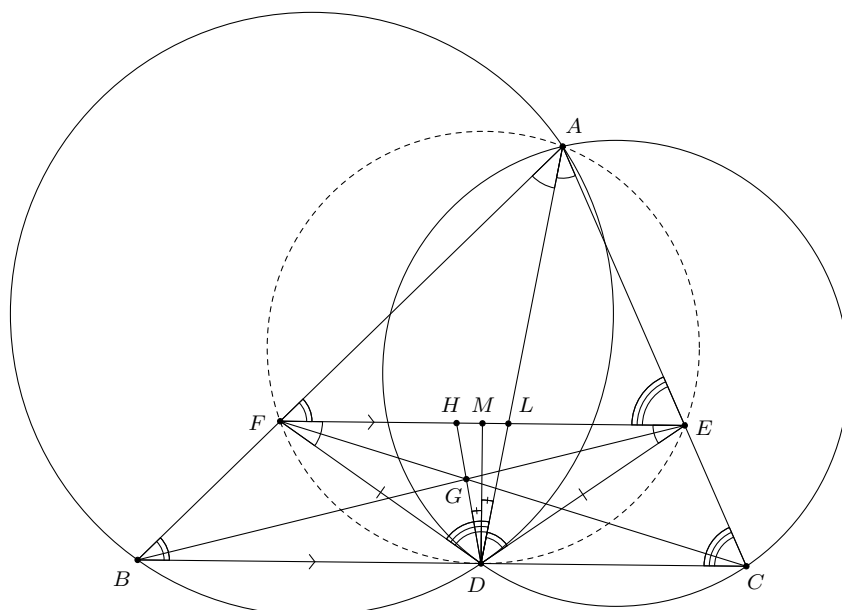
Since $\widehat{AB}$ not containing C on the circumcircle of $\triangle ABC$, we have $BM = AM$. Now observe that

- $\angle MBD = \angle MAE$ (because $AMBC$ is cyclic);
- $\angle MDB = \angle MFA$ (because $MDBF$ is cyclic); and
- $\angle MFA = \angle MEA$ (because $AMFE$ is cyclic).

Hence $\angle MDB = \angle MEA$. Therefore $\triangle BMD \cong \triangle AME$ and so $BD = AE$. □

- Problem 58.** In $\triangle ABC$, the internal angle bisector of $\angle BAC$ meets side BC at D . The lines through D tangents to the circumcircles of $\triangle ABD$ and $\triangle ACD$ meet lines AC and AB at points E and F , respectively. Lines BE and CF intersect at G . Prove that $\angle EDG = \angle ADF$.

Solution 1.



First observe that because of tangency at D , we have $\angle EDA = \angle ABC$ and $\angle ADF = \angle BCA$. Consequently

$$\angle EDF + \angle FAE = (\angle EDA + \angle ADF) + \angle BAC = \angle ABC + \angle BCA + \angle BAC = 180^\circ,$$

and so A, F, D, E are concyclic, which in turn implies that

$$\angle AFE = \angle ADE = \angle ABC \quad \text{and} \quad \angle AEF = \angle ADF = \angle BCA.$$

Hence $EF \parallel BC$. Moreover, $\triangle FDE$ is isosceles because

$$\angle EFD = \angle EAD = \angle FAD = \angle FED.$$

Now let M be the midpoint of EF , and let $L = AD \cap EF$, and $H = DG \cap EF$. By the Intercept Theorem,

$$\frac{LE}{LF} = \frac{DC}{DB}. \quad (*)$$

Moreover, since $\triangle FGH \sim \triangle CGD$, we have

$$\frac{HF}{DC} = \frac{GH}{GD}.$$

Since $\triangle EGH \sim \triangle BGD$, we also have

$$\frac{HE}{DB} = \frac{GH}{GD}$$

and so

$$\frac{HF}{HE} = \frac{DC}{DB}.$$

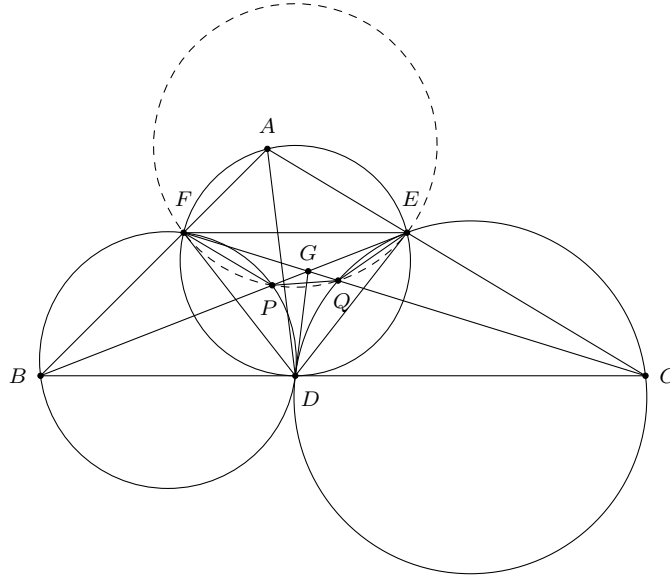
Since $LF + LE = EF = HF + HE$, we have

$$\frac{EF}{LF} = 1 + \frac{LE}{LF} = 1 + \frac{HF}{HE} = \frac{EF}{HE}$$

and so $LF = HE$, and $LE = HF$. Hence $HM = ML$. Therefore $\angle HDM = \angle LDM$, and so

$$\angle EDG = 90^\circ - \angle FED + \angle HDM = 90^\circ - \angle EFD + \angle LDM = \angle ADF. \quad \square$$

Solution 2.



Since DE is tangent to (ABD) , we have $\angle ADE = \angle ABD$. As $\angle DAE = \angle BAD$, we have $\triangle ADE \sim \triangle ABD$. Therefore, BD is tangent to (ADE) . Similarly, CD is tangent to (ADF) . This shows A, F, D, E are concyclic. As AD bisects $\angle FAE$, we have $DE = DF$. Also, note that $\angle AFE = \angle ADE = \angle ABD$, so that $FE \parallel BC$.

Now, from $\angle EDF = \angle ADE + \angle ADF = \angle FBD + \angle ECD$, the circles (BDF) and (CDE) are tangent to each other at D . Let (BDF) meet BE at P , and let (CDE) meet CF at Q . From $\angle FEP = \angle DBP = \angle DFP$, we know that DF is a tangent to (EFP) . Similarly, DE is a tangent to (EFQ) . If (EFP) and (EFQ) are different circles, then because $DF = DE$, the powers of D with respect to the two circles are the same and so D lies on their radical axis. But their radical axis is obviously EF and clearly does not pass through D . This is a contradiction. Therefore, E, F, P, Q are concyclic.

It follows that $\angle GPQ = \angle EFQ = \angle BCQ$. This implies B, C, Q, P are concyclic. Then $GB \times GP = GC \times GQ$, hence G lies on the radical axis of (BDF) and (CDE) . Therefore, DG is the common tangent to the two circles. Then $\angle EDG = \angle ECD = \angle ADF$. $\square$

Problem 59. An *admissible* expression of a positive integer is an expression as a sum of powers of 3, 4 and 7 in such a way that the expression does not contain two powers with the same base and the same exponent.

For example, $2 = 7^0 + 7^0$ and $22 = 3^2 + 3^2 + 4^1$ are not admissible expressions, while $2 = 3^0 + 7^0$ and $22 = 3^2 + 3^0 + 4^1 + 4^0 + 7^1$ are admissible expressions.

Prove that every positive integer has an admissible expression.

Solution. Let $x_1, x_2, x_3, \dots \in \{3, 4, 7\}$ and let $a_1, a_2, a_3, \dots$ be non-negative integers such that $x_1^{a_1} \leq x_2^{a_2} \leq x_3^{a_3} \leq \dots$ and the list contains all powers of 3, 4, 7. We will prove by induction on n that all the integers from 1 to $x_1^{a_1} + x_2^{a_2} + \dots + x_n^{a_n}$ have admissible expressions using only the powers $x_1^{a_1}, x_2^{a_2}, \dots, x_n^{a_n}$. The base cases $n = 1, 2, 3$ hold because $1 = 3^0, 2 = 3^0 + 4^0, 3 = 3^0 + 4^0 + 7^0$. Assuming the property holds for $n \geq 3$, we will prove that it is valid for $n + 1$.

If the powers $x_1^{a_1}, x_2^{a_2}, \dots, x_n^{a_n}$ are $\{3^0, 3^1, \dots, 3^a\} \cup \{4^0, 4^1, \dots, 4^b\} \cup \{7^0, 7^1, \dots, 7^c\}$, for some non-negative integers a, b, c , we have

$$\begin{aligned} x_1^{a_1} + x_2^{a_2} + \dots + x_n^{a_n} &= (3^0 + 3^1 + \dots + 3^a) + (4^0 + 4^1 + \dots + 4^b) + (7^0 + 7^1 + \dots + 7^c) \\ &= \frac{3^{a+1} - 1}{2} + \frac{4^{b+1} - 1}{3} + \frac{7^{c+1} - 1}{6} \\ &\geq \frac{x_{n+1}^{a_{n+1}} - 1}{2} + \frac{x_{n+1}^{a_{n+1}} - 1}{3} + \frac{x_{n+1}^{a_{n+1}} - 1}{6} \\ &= x_{n+1}^{a_{n+1}} - 1. \end{aligned}$$

Then by the induction assumption, all the positive integers smaller than $x_{n+1}^{a_{n+1}}$ have admissible expressions using only the powers $x_1^{a_1}, x_2^{a_2}, \dots, x_{n+1}^{a_{n+1}}$. Moreover, an integer m such that

$$x_{n+1}^{a_{n+1}} \leq m \leq x_1^{a_1} + x_2^{a_2} + \dots + x_n^{a_n} + x_{n+1}^{a_{n+1}}$$

can be written as

$$m = (m - x_{n+1}^{a_{n+1}}) + x_{n+1}^{a_{n+1}}, \text{ where } 0 \leq m - x_{n+1}^{a_{n+1}} \leq x_1^{a_1} + x_2^{a_2} + \dots + x_n^{a_n},$$

and so has an admissible expression using only $x_1^{a_1}, x_2^{a_2}, \dots, x_n^{a_n}$. We conclude that all integers from 1 to $x_1^{a_1} + x_2^{a_2} + \dots + x_{n+1}^{a_{n+1}}$ have admissible expressions using only the powers $x_1^{a_1}, x_2^{a_2}, \dots, x_n^{a_n}, x_{n+1}^{a_{n+1}}$, which completes the induction. $\square$

Problem 60. A convex 2020-gon is given for which no three diagonals intersect in a point. The intersection point of two internal diagonals of the given polygon will be called a *knot*. Two knots are called *neighbours* if they share a common diagonal. A closed path between neighbouring knots such that no three consecutive knots share a diagonal will be called a *cycle*.

Find the maximal number of knots that can be coloured such that there exists no coloured cycle, that is, a cycle containing only coloured knots?

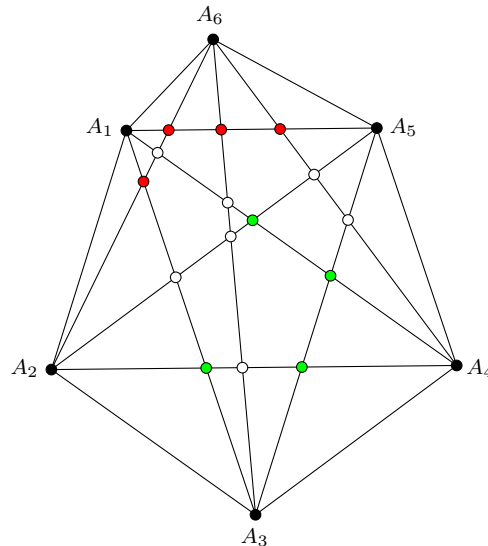
Solution. The answer is $\frac{2020(2020-3)}{2} - 1 = 2037169$.

We will prove that for a convex n -gon with $n \geq 4$ satisfying the problem condition, the maximal number is given by $\frac{n(n-3)}{2} - 1$.

First, we show that the number of coloured knots needs to be less than $\frac{n(n-3)}{2}$. Assume the contrary and consider a graph, with vertices corresponding to the internal diagonals of the polygon and with edges corresponding to the coloured knots. Since the number of diagonals in a convex n -gon equals $\binom{n}{2} - n = \frac{n(n-3)}{2}$, the edges in the graph outnumber the vertices, and so the graph contains a cycle. But it is easy to see that there is a one-to-one correspondence between the cycles in the graph and the coloured cycles in the polygon. Hence, there exists a coloured cycle, which is a contradiction.

Next we show a strategy for colouring $\frac{n(n-3)}{2} - 1$ knots such that no coloured cycle exists. We do this by induction. The base case $n = 4$ is trivial. For the inductive step, consider a convex $(n+1)$ -gon $A_1 A_2 \dots A_{n+1}$ and take a colouring of the knots in $A_1 A_2 \dots A_n$ satisfying the problem condition. By adding the vertex A_{n+1} , we transform the side $A_1 A_n$ of $A_1 A_2 \dots A_n$ into an internal diagonal of $A_1 A_2 \dots A_{n+1}$ and we obtain $n-2$ internal diagonals $A_{n+1} A_i$, where $i = 2, \dots, n-1$ more. Hence the internal diagonal $A_1 A_n$ contains exactly $n-2$ knots (all corresponding to the additional internal diagonals). We colour all of them. Finally, we colour one more knot on an internal diagonal through A_{n+1} , say the intersection point of $A_{n+1} A_2$ and $A_1 A_3$. Note that due to the convexity of the polygon, it is clear that all considered intersection points are internal, and so are indeed knots!

The figure below shows the inductive step from a pentagon to a hexagon where the green knots correspond to the pentagon and the red knots are the additional coloured knots.



In total, we end up with

$$\frac{n(n-3)}{2} - 1 + (n-1) = \frac{(n+1)(n-2)}{2} - 1$$

coloured knots. Among the $n-1$ additionally coloured knots, coloured neighbours occur only in the internal diagonals $A_1 A_n$ or $A_2 A_{n+1}$. Furthermore, every other internal diagonal through A_{n+1} contains at most one coloured knot, i.e. a coloured knot has not coloured neighbour. Now observe that in order for a coloured cycle to exist given that the colouring of the knots of $A_1 A_2 \dots A_n$ satisfy the problem condition, at least two of the additionally coloured knots should be part of this cycle. This is impossible due to construction and so completes the induction and the proof of the required number of maximal coloured knots. $\square$